

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT II (BJT AMPLIFIERS)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.V.SrirengaNachiyar

Name of the Topic: Cascade, Cascode configurations

Name of the Innovative Practice: Simulation tool (MULTISIM)

Date & Duration: 01.08.2019 & 20 Minutes

ICT Tool used: Multisim

Description:

Simulation:

Simulation is the imitation of the operation of a real-world process or system over time. The act of simulating something first requires that a model be developed; this model represents the key characteristics or behaviors of the selected physical or abstract system or process. The model represents the system itself, whereas the simulation represents the operation of the system over time.

Learning Outcome:

The students will be able to understand the concepts of cascade and cascode configurations clearly by using simulation tool.

Use of appropriate method:

Justification for choosing the topic using simulation activity:

Simulation activity shows the waveform which consists of frequency, bandwidth and gain of the cascade and cascode amplifiers. It will be helpful for understanding and differentiate the operation and gain of an amplifier.

Effective Presentation:

I have planned this activity for 20 minutes, but students took 30 minutes to complete. Because they don't know about the simulation tool. So, I explained the simulation tool, components and execution of the circuit. Then students complete this activity with interest.

Significance of Results:

The Students eagerly participated in that activity and then they acquired the concept of cascade and cascode amplifier configurations.

Reflective Critique:

Challenges:

The students couldn't able to understand, how to work with simulation tool. They struggled to simulate the configurations. After that, I have explained the steps to be followed for simulation.

Benefits:

I have chosen the simulation tool for this topic for visualize the gain and bandwidth for differentiating cascade and cascode amplifier configurations. The students eagerly participated in that activity and then they acquired the concept of cascade and cascade amplifier configurations.

Innovative Teaching Method Execution



Changes required for future (if required):

Before start doing this activity, first clearly explained about this simulation tool, components and execution of the circuit to the students.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	2	2	2	2	3	-	-	-	2	3	-	2

CO2: The students will be able to analyze the small signal equivalent and design BJT amplifier circuits

References:

1. David A. Bell, Electronic Devices & Circuits
2. https://www.google.com/search?q=multisim+simulation&rlz=1C1GCEU_enIN844IN844&oq=Multisim+simulation&aqs=chrome.0.0l8.8591j0j7&sourceid=chrome&ie=UTF-8

RAMCO INSTITUTE OF TECHNOLOGY
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Innovative Practices Description

UNIT III (SINGLE STAGE FET, MOSFET AMPLIFIERS)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.V.SrIREngaNachiyar

Name of the Topic: BiCMOS Circuits

Name of the Innovative Practice: Mind map

Date & Duration: 06.09.2019 & 20 Minutes

ICT Tool used: Whiteboard

Description:

Mind Map Technique:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall. A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Learning Outcomes:

1. The students can be acquired better knowledge about BiCMOS Circuits.
2. The students understood the advantages of the combination of BJT and MOSFET called BiCMOS circuits compared to BJT and MOSFET separately.

Use of appropriate method:

Justification for choosing the Mind Map Activity:

Mind map activity gives the graphical representation of BiCMOS circuits. It explains the concepts of BiCMOS circuits clearly which will be helpful for understanding and able to differentiate BiCMOS circuit with BJT and MOSFET.

Effective Presentation:

Initially, I have planned this activity for 20 minutes, but students took 30 minutes to complete. Because they don't know about this activity and how to draw the map for this topic. So, I explained, then students able to complete this activity with interest.

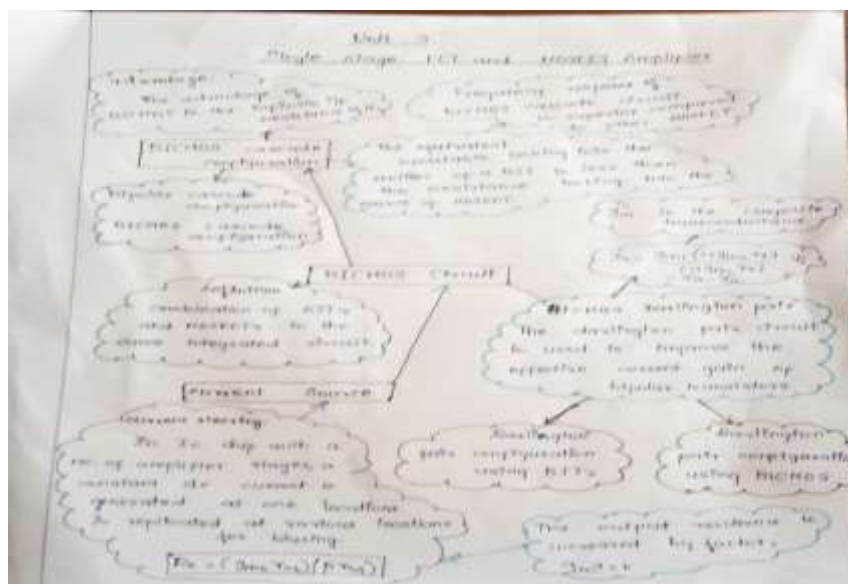
Significance of Results:

The Students eagerly participated in this activity and then they acquired the concept of BiCMOS amplifier configurations.

Challenges:

Benefits:

BiCMOS Circuits:- Mind Map



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	1	2	2	1						2	2

CO3: The students will be able to analyze the small signal equivalent and design JFET and MOSFET amplifier circuits.

References:

1. Salivahanan and N. Suresh Kumar, Electronic Devices and Circuits
2. <https://simplemind.eu/how-to-mind-map/basics/>

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Innovative Practices Description

UNIT IV (FREQUENCY RESPONSE OF AMPLIFIERS)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.V.SrirengaNachiyar

Name of the Topic: Frequency response of amplifiers

Name of the Innovative Practice: Brainstorming

Date & Duration: 14.09.2019 & 15 Minutes

ICT tool used: Whiteboard

Description:

Brainstorming:

Brainstorming is a method for generating ideas to solve a design problem. It usually involves a group, under the direction of a facilitator. The strength of brainstorming is the potential participants have in drawing associations between their ideas in a free-thinking environment, thereby broadening the solution space.

The normal view of brainstorming is where a group of people sit in a room and shout out ideas as they occur to them. They are told to lose their inhibitions and that no ideas will be judged so that people are free to shout out any ideas at all without feeling uncomfortable. People should build on the ideas called out by other participants. The purpose of this is to gain as many ideas as possible for later analysis. Out of the many ideas suggested there will be some of great value. Because of the free-thinking environment, the session will help promote radical new ideas which break free from normal ways of thinking.

Learning Outcomes:

1. The students can be acquired better knowledge about Frequency response of an amplifier.
2. The students understood the 3dB response of BJT.

Use of appropriate method:

Justification for choosing Brainstorming Activity:

Before the topic of frequency response of an amplifier, I have taught general response of an amplifier. Therefore, students should know about response, -3dB frequency. For that reason, I have chosen Brainstorming activity for this topic. Definitely, it helps to understand the -3dB frequency and frequency response of an amplifier.

Effective Presentation:

I have planned the activity for 10 minutes but students struggled to recall about -3dB frequency and then I gave another 5 minutes to recall it. Totally, it took 15 minutes to complete it. Then students complete this activity with interest

Reflective Critique:

Challenges:

Some of the students did mistakes while doing the activity. It takes more time to recall about frequency response which was taught in previous lectures. After that, students eagerly participated and brainstormed their views. However, some of them did mistake but most of them explained correctly. At last, I concluded about the frequency response of amplifiers.

Benefits:

I have chosen the brainstorming for this topic for gaining more knowledge about the frequency response of amplifiers. The students eagerly participated in that activity and then they acquired the better knowledge about frequency response of an amplifier and -3db frequency.

Innovative Teaching Method Execution	
Frequency response of amplifiers	
	
	

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	2	2	2	2	1				2		2	2

CO4: The students will be able to plot the frequency response of all amplifiers

References:

1. Donald. A. Neamen: Electronic Circuits Analysis and Design
2. <https://en.wikipedia.org/wiki/Brainstorming>

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UNIT V (POWER SUPPLIES AND ELECTRONIC DEVICE TESTING)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8351 Electronic Circuits I

Name of the Faculty member: Mrs.V.SrirengaNachiyar

Name of the Topic: Rectifiers

Name of the Innovative Practice: Role Play

Date & Duration: 25.09.2019 & 15 Minutes

Description:

Role Play Technique:

Role-play is a technique that allows students to explore realistic situations by interacting with other people in a managed way in order to develop experience and trial different strategies in a supported environment.

Role play is an educational technique in which people spontaneously act-out problems of human relations and analyze the enactment with the help of other role players and observers.

Role playing is effective when the topic involves person to person communication or interactions. It can allow everyone to participate. The overall mood a training session can be improved by the excitement that role playing provide.

Learning Outcomes:

1. The students can be acquired the better knowledge about half wave and full wave rectifiers.
2. The students can be easily differentiated the working principles of half wave and full wave rectifiers.

Use of appropriate method:

Justification for choosing the Role play Activity:

Role play activity gives the concept of rectifiers elaborately. This activity helps the students to understand the working of each and every component present in the rectifier circuit.

Effective Presentation:

I have planned this activity for 15 minutes, but students took 20 minutes to complete. Because they don't know about this activity. So, I explained the activity, then students enjoy to do this activity with interest.

Reflective Critique:

Challenges:

The students confused the working principle of half wave and full wave rectifiers. Because the two rectifiers works under same principle with small difference. After that, I have chosen role play as an Innovative practice method for better understanding about the rectifiers. The Students eagerly participated in that activity and then they acquired the concept of rectifiers.

Benefits:

Because of this activity, the students can be able to understand the concepts clearly and differentiate the working principle of the rectifiers.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	2	2	2	2					2	2		2

CO5: The students will be able to design the regulated power supply, troubleshoot and analyze the faults in power supplies.

References:

1. https://www.google.com/search?safe=strict&ei=RujkXfHzM7uQ4-EPkb6ouAw&q=importance+of+role+play&oq=role+play&gs_l=psy-ab.1.0.0i7118.0.0..16250...0.2..0.0.0.....0.....gws-wiz.GZGnSO2X2PY